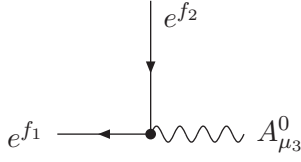
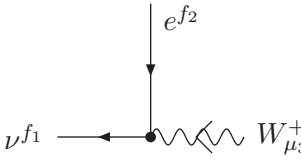
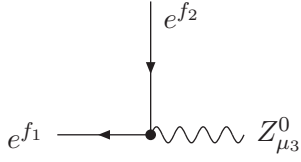
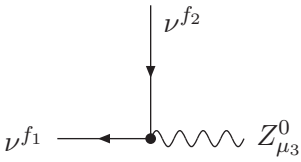


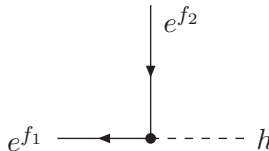
A SMEFT interaction vertices

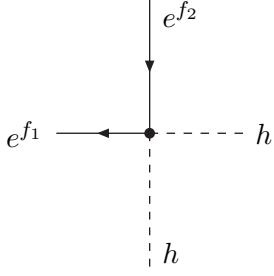
CAUTION: interaction vertices printed below are shown including only terms up to mass dimension-6. Interactions proportional to products of dimension-6 Wilson coefficients, even if calculated and included in other output formats (Mathematica, Feynarts, UFO etc.), are too complicated for printout and for manual calculations. If necessary, they can be inspected visually displaying relevant variables (for their list see `SmeftFR` manual) in the Mathematica notebook.

A.1 Lepton–gauge vertices

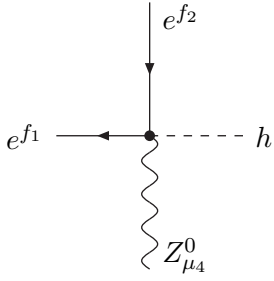
	$+ \frac{i\bar{g}'\bar{g}}{\sqrt{\bar{g}'^2 + \bar{g}^2}} \delta_{f_1 f_2} \gamma^{\mu_3} - \frac{i\bar{g}'^2 \bar{g}^2 v^2}{(\bar{g}'^2 + \bar{g}^2)^{3/2}} \delta_{f_1 f_2} C^{\varphi WB} \gamma^{\mu_3}$
	$- \frac{i\bar{g}}{\sqrt{2}} U_{f_2 f_1}^* \gamma^{\mu_3} P_L$
	$\begin{aligned} & - \frac{i}{2\sqrt{\bar{g}'^2 + \bar{g}^2}} \delta_{f_1 f_2} ((\bar{g}'^2 - \bar{g}^2) \gamma^{\mu_3} P_L + 2\bar{g}'^2 \gamma^{\mu_3} P_R) \\ & + \frac{i\bar{g}'\bar{g}v^2}{2(\bar{g}'^2 + \bar{g}^2)^{3/2}} \delta_{f_1 f_2} C^{\varphi WB} ((\bar{g}'^2 - \bar{g}^2) \gamma^{\mu_3} P_L - 2\bar{g}^2 \gamma^{\mu_3} P_R) \\ & + \frac{1}{2} i v^2 \sqrt{\bar{g}'^2 + \bar{g}^2} C_{f_1 f_2}^{\varphi l1} \gamma^{\mu_3} P_L \end{aligned}$
	$\begin{aligned} & - \frac{1}{2} i \sqrt{\bar{g}'^2 + \bar{g}^2} \delta_{f_1 f_2} \gamma^{\mu_3} P_L - \frac{i\bar{g}'\bar{g}v^2}{2\sqrt{\bar{g}'^2 + \bar{g}^2}} \delta_{f_1 f_2} C^{\varphi WB} \gamma^{\mu_3} P_L \\ & + \frac{1}{2} i v^2 \sqrt{\bar{g}'^2 + \bar{g}^2} U_{g_2 f_2} U_{g_1 f_1}^* C_{g_1 g_2}^{\varphi l1} \gamma^{\mu_3} P_L \end{aligned}$

A.2 Lepton–Higgs–gauge vertices

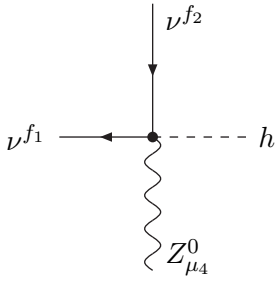
	$- \frac{i}{v} m_{l_{f_1}} \delta_{f_1 f_2} + \frac{i v^2}{\sqrt{2}} \left(P_L C_{f_2 f_1}^{e\varphi*} + P_R C_{f_1 f_2}^{e\varphi} \right)$
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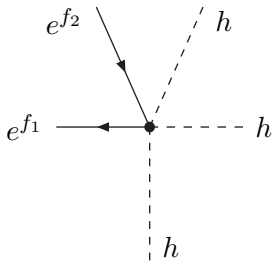
$$+\frac{3iv}{\sqrt{2}}\left(P_L C_{f_2 f_1}^{e\varphi*} + P_R C_{f_1 f_2}^{e\varphi}\right)$$



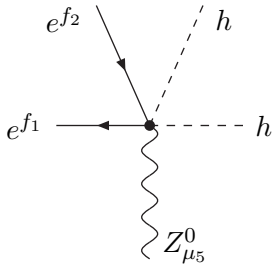
$$+iv\sqrt{\bar{g}'^2 + \bar{g}^2} C_{f_1 f_2}^{\varphi l1} \gamma^{\mu_4} P_L$$



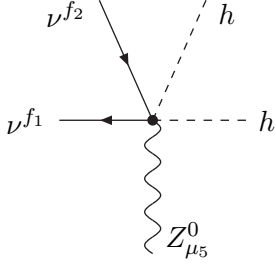
$$+iv\sqrt{\bar{g}'^2 + \bar{g}^2} U_{g_2 f_2} U_{g_1 f_1}^* C_{g_1 g_2}^{\varphi l1} \gamma^{\mu_4} P_L$$



$$+\frac{3i}{\sqrt{2}}\left(P_L C_{f_2 f_1}^{e\varphi*} + P_R C_{f_1 f_2}^{e\varphi}\right)$$

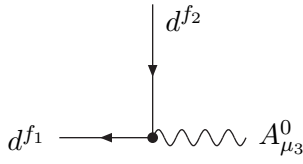


$$+i\sqrt{\bar{g}'^2 + \bar{g}^2} C_{f_1 f_2}^{\varphi l1} \gamma^{\mu_5} P_L$$

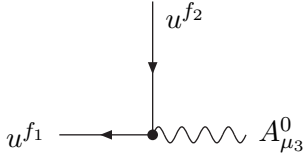


$$+i\sqrt{\bar{g}'^2 + \bar{g}^2} U_{g_2 f_2} U_{g_1 f_1}^* C_{g_1 g_2}^{\varphi l 1} \gamma^{\mu_5} P_L$$

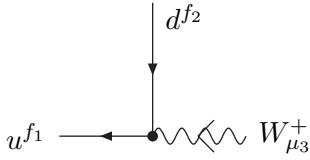
A.3 Quark-gauge vertices



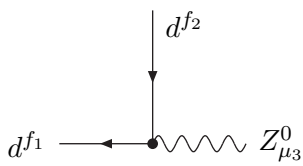
$$+ \frac{i\bar{g}'\bar{g}}{3\sqrt{\bar{g}'^2 + \bar{g}^2}} \delta_{f_1 f_2} \gamma^{\mu_3} - \frac{i\bar{g}'^2 \bar{g}^2 v^2}{3(\bar{g}'^2 + \bar{g}^2)^{3/2}} \delta_{f_1 f_2} C^{\varphi WB} \gamma^{\mu_3} \\ + \frac{\sqrt{2}\bar{g}v}{\sqrt{\bar{g}'^2 + \bar{g}^2}} p_3^\nu \left(C_{f_2 f_1}^{dB*} \sigma^{\nu \mu_3} P_L + C_{f_1 f_2}^{dB} \sigma^{\nu \mu_3} P_R \right)$$



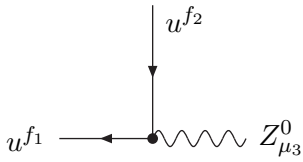
$$- \frac{2i\bar{g}'\bar{g}}{3\sqrt{\bar{g}'^2 + \bar{g}^2}} \delta_{f_1 f_2} \gamma^{\mu_3} + \frac{2i\bar{g}'^2 \bar{g}^2 v^2}{3(\bar{g}'^2 + \bar{g}^2)^{3/2}} \delta_{f_1 f_2} C^{\varphi WB} \gamma^{\mu_3}$$



$$- \frac{i\bar{g}}{\sqrt{2}} K_{f_1 f_2} \gamma^{\mu_3} P_L$$

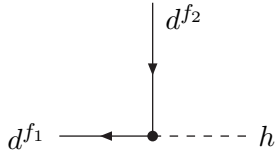


$$+ \frac{i}{6\sqrt{\bar{g}'^2 + \bar{g}^2}} \delta_{f_1 f_2} ((\bar{g}'^2 + 3\bar{g}^2) \gamma^{\mu_3} P_L - 2\bar{g}'^2 \gamma^{\mu_3} P_R) \\ + \frac{i\bar{g}'\bar{g}v^2}{6(\bar{g}'^2 + \bar{g}^2)^{3/2}} \delta_{f_1 f_2} C^{\varphi WB} ((3\bar{g}'^2 + \bar{g}^2) \gamma^{\mu_3} P_L - 2\bar{g}^2 \gamma^{\mu_3} P_R) \\ - \frac{\sqrt{2}\bar{g}'v}{\sqrt{\bar{g}'^2 + \bar{g}^2}} p_3^\nu \left(C_{f_2 f_1}^{dB*} \sigma^{\nu \mu_3} P_L + C_{f_1 f_2}^{dB} \sigma^{\nu \mu_3} P_R \right)$$

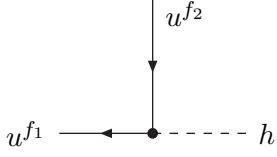


$$+ \frac{i}{6\sqrt{\bar{g}'^2 + \bar{g}^2}} \delta_{f_1 f_2} ((\bar{g}'^2 - 3\bar{g}^2) \gamma^{\mu_3} P_L + 4\bar{g}'^2 \gamma^{\mu_3} P_R) \\ - \frac{i\bar{g}'\bar{g}v^2}{6(\bar{g}'^2 + \bar{g}^2)^{3/2}} \delta_{f_1 f_2} C^{\varphi WB} ((3\bar{g}'^2 - \bar{g}^2) \gamma^{\mu_3} P_L - 4\bar{g}^2 \gamma^{\mu_3} P_R)$$

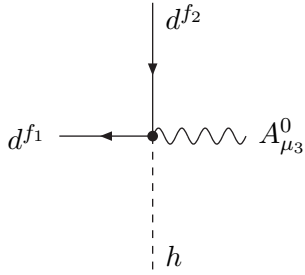
A.4 Quark-Higgs-gauge vertices



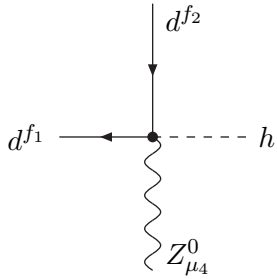
$$-\frac{i}{v}m_{d_{f_1}}\delta_{f_1 f_2}$$



$$-\frac{i}{v}m_{u_{f_1}}\delta_{f_1 f_2}$$

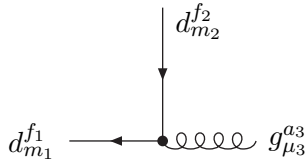


$$+\frac{\sqrt{2}\bar{g}}{\sqrt{\bar{g}'^2+\bar{g}^2}}p_3^\nu\left(C_{f_2 f_1}^{dB*}\sigma^{\nu\mu_3}P_L+C_{f_1 f_2}^{dB}\sigma^{\nu\mu_3}P_R\right)$$

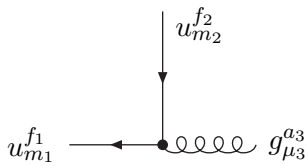


$$-\frac{\sqrt{2}\bar{g}'}{\sqrt{\bar{g}'^2+\bar{g}^2}}p_4^\nu\left(C_{f_2 f_1}^{dB*}\sigma^{\nu\mu_4}P_L+C_{f_1 f_2}^{dB}\sigma^{\nu\mu_4}P_R\right)$$

A.5 Quark-gluon vertices

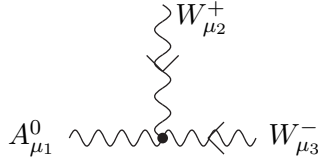


$$-i\bar{g}_s\delta_{f_1 f_2}\mathcal{T}_{m_1 m_2}^{a_3}\gamma^{\mu_3}$$

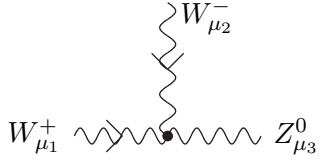


$$-i\bar{g}_s\delta_{f_1 f_2}\mathcal{T}_{m_1 m_2}^{a_3}\gamma^{\mu_3}$$

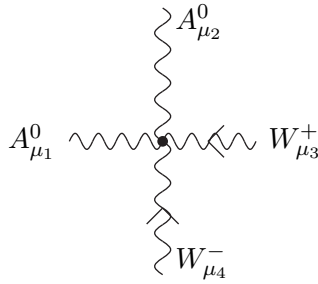
A.6 Gauge self interaction vertices



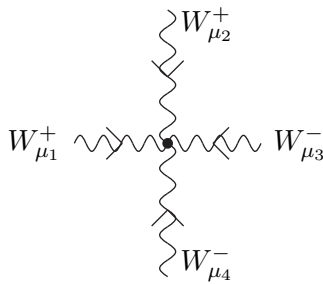
$$\begin{aligned}
& + \frac{i\bar{g}'\bar{g}}{\sqrt{\bar{g}'^2 + \bar{g}^2}} (\eta_{\mu_1\mu_2}p_1^{\mu_3} - \eta_{\mu_1\mu_2}p_2^{\mu_3} - \eta_{\mu_1\mu_3}p_1^{\mu_2} + \eta_{\mu_1\mu_3}p_3^{\mu_2} + \eta_{\mu_2\mu_3}p_2^{\mu_1} \\
& - \eta_{\mu_2\mu_3}p_3^{\mu_1}) + \frac{i\bar{g}^2v^2}{(\bar{g}'^2 + \bar{g}^2)^{3/2}} C^{\varphi WB} (\bar{g}'^2\eta_{\mu_1\mu_2}p_2^{\mu_3} - \bar{g}'^2\eta_{\mu_1\mu_3}p_3^{\mu_2} \\
& - \bar{g}'^2\eta_{\mu_2\mu_3}p_2^{\mu_1} + \bar{g}'^2\eta_{\mu_2\mu_3}p_3^{\mu_1} + \bar{g}^2\eta_{\mu_1\mu_2}p_1^{\mu_3} - \bar{g}^2\eta_{\mu_1\mu_3}p_1^{\mu_2})
\end{aligned}$$



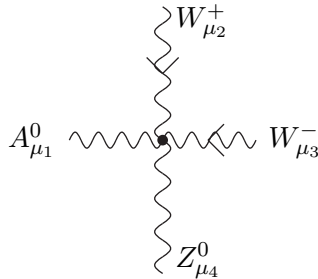
$$\begin{aligned}
& + \frac{i\bar{g}^2}{\sqrt{\bar{g}'^2 + \bar{g}^2}} (\eta_{\mu_1\mu_2}p_1^{\mu_3} - \eta_{\mu_1\mu_2}p_2^{\mu_3} - \eta_{\mu_1\mu_3}p_1^{\mu_2} + \eta_{\mu_1\mu_3}p_3^{\mu_2} + \eta_{\mu_2\mu_3}p_2^{\mu_1} \\
& - \eta_{\mu_2\mu_3}p_3^{\mu_1}) + \frac{i\bar{g}'\bar{g}v^2}{(\bar{g}'^2 + \bar{g}^2)^{3/2}} C^{\varphi WB} (\bar{g}'^2\eta_{\mu_1\mu_2}p_1^{\mu_3} - \bar{g}'^2\eta_{\mu_1\mu_2}p_2^{\mu_3} \\
& - \bar{g}'^2\eta_{\mu_1\mu_3}p_1^{\mu_2} + \bar{g}'^2\eta_{\mu_2\mu_3}p_2^{\mu_1} - \bar{g}^2\eta_{\mu_1\mu_3}p_3^{\mu_2} + \bar{g}^2\eta_{\mu_2\mu_3}p_3^{\mu_1})
\end{aligned}$$



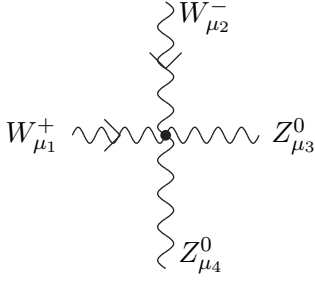
$$\begin{aligned}
& + \frac{i\bar{g}'^2\bar{g}^2}{\bar{g}'^2 + \bar{g}^2} (\eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3} + \eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} - 2\eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4}) \\
& - \frac{2i\bar{g}'^3\bar{g}^3v^2}{(\bar{g}'^2 + \bar{g}^2)^2} (\eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3} + \eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} - 2\eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4}) C^{\varphi WB}
\end{aligned}$$



$$-i\bar{g}^2 (\eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3} + \eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} - 2\eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4})$$

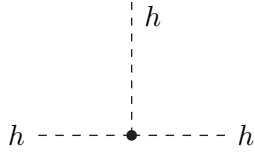


$$\begin{aligned}
& - \frac{i\bar{g}'\bar{g}^3}{\bar{g}'^2 + \bar{g}^2} (2\eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3} - \eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} - \eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4}) - \frac{i\bar{g}'^2\bar{g}^2v^2}{(\bar{g}'^2 + \bar{g}^2)^2} (\bar{g}' \\
& - \bar{g})(\bar{g}' + \bar{g}) (2\eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3} - \eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} - \eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4}) C^{\varphi WB}
\end{aligned}$$

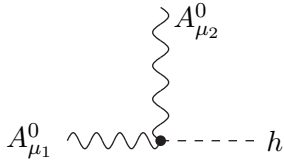


$$\begin{aligned}
& + \frac{i\bar{g}^4}{\bar{g}'^2 + \bar{g}^2} (\eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3} + \eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} - 2\eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4}) \\
& + \frac{2i\bar{g}'^3\bar{g}^3v^2}{(\bar{g}'^2 + \bar{g}^2)^2} (\eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3} + \eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} - 2\eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4}) C^{\varphi WB}
\end{aligned}$$

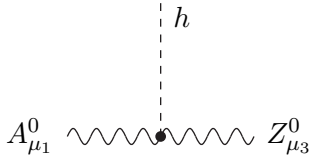
A.7 Higgs-gauge vertices



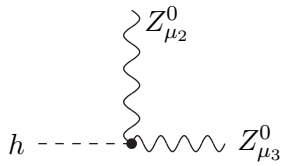
$$-3i\lambda v$$



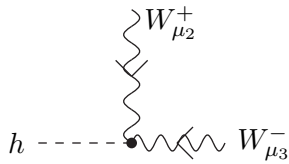
$$-\frac{4i\bar{g}'\bar{g}v}{\bar{g}'^2 + \bar{g}^2} C^{\varphi WB} (p_1^{\mu_2} p_2^{\mu_1} - p_1 \cdot p_2 \eta_{\mu_1\mu_2})$$



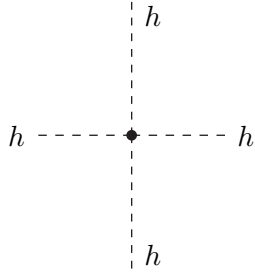
$$+\frac{2iv}{\bar{g}'^2 + \bar{g}^2} (\bar{g}' - \bar{g})(\bar{g}' + \bar{g}) C^{\varphi WB} (p_1^{\mu_3} p_3^{\mu_1} - p_1 \cdot p_3 \eta_{\mu_1\mu_3})$$



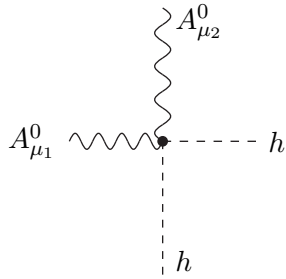
$$\begin{aligned}
& + \frac{iv}{2} (\bar{g}'^2 + \bar{g}^2) \eta_{\mu_2\mu_3} \\
& + \frac{i\bar{g}'\bar{g}v}{\bar{g}'^2 + \bar{g}^2} C^{\varphi WB} (\eta_{\mu_2\mu_3} (-4p_2 \cdot p_3 + \bar{g}'^2 v^2 + \bar{g}^2 v^2) + 4p_2^{\mu_3} p_3^{\mu_2})
\end{aligned}$$



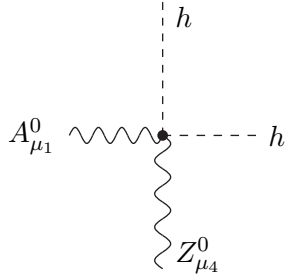
$$+\frac{1}{2}i\bar{g}^2 v \eta_{\mu_2\mu_3}$$



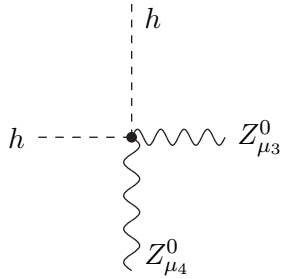
$$-3i\lambda$$



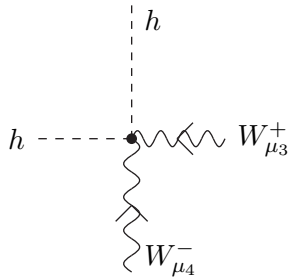
$$-\frac{4i\bar{g}'\bar{g}}{\bar{g}'^2 + \bar{g}^2} C^{\varphi WB} (p_1^{\mu_2} p_2^{\mu_1} - p_1 \cdot p_2 \eta_{\mu_1 \mu_2})$$



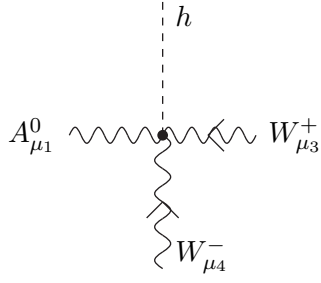
$$+\frac{2i}{\bar{g}'^2 + \bar{g}^2} (\bar{g}' - \bar{g})(\bar{g}' + \bar{g}) C^{\varphi WB} (p_1^{\mu_4} p_4^{\mu_1} - p_1 \cdot p_4 \eta_{\mu_1 \mu_4})$$



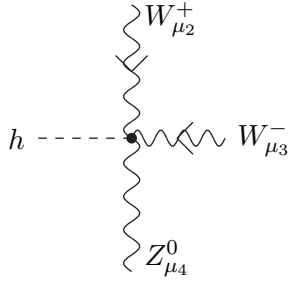
$$+\frac{i}{2} (\bar{g}'^2 + \bar{g}^2) \eta_{\mu_3 \mu_4} + \frac{i\bar{g}'\bar{g}}{\bar{g}'^2 + \bar{g}^2} C^{\varphi WB} (\eta_{\mu_3 \mu_4} (-4p_3 \cdot p_4 + \bar{g}'^2 v^2 + \bar{g}^2 v^2) + 4p_3^{\mu_4} p_4^{\mu_3})$$



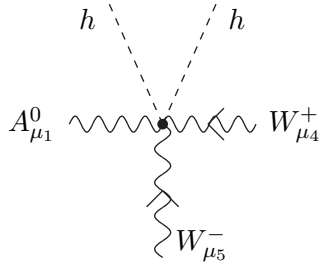
$$+\frac{i\bar{g}^2}{2} \eta_{\mu_3 \mu_4}$$



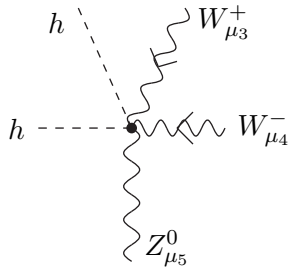
$$+ \frac{2i\bar{g}^2 v}{\sqrt{\bar{g}'^2 + \bar{g}^2}} C^{\varphi WB} (\eta_{\mu_1 \mu_3} p_1^{\mu_4} - \eta_{\mu_1 \mu_4} p_1^{\mu_3})$$



$$- \frac{2i\bar{g}'\bar{g}v}{\sqrt{\bar{g}'^2 + \bar{g}^2}} C^{\varphi WB} (\eta_{\mu_2 \mu_4} p_4^{\mu_3} - \eta_{\mu_3 \mu_4} p_4^{\mu_2})$$

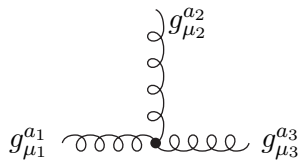


$$+ \frac{2i\bar{g}^2}{\sqrt{\bar{g}'^2 + \bar{g}^2}} C^{\varphi WB} (\eta_{\mu_1 \mu_4} p_1^{\mu_5} - \eta_{\mu_1 \mu_5} p_1^{\mu_4})$$

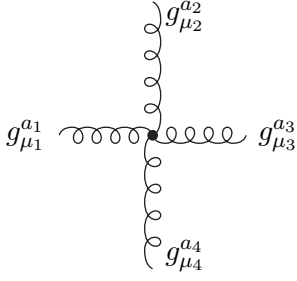


$$- \frac{2i\bar{g}'\bar{g}}{\sqrt{\bar{g}'^2 + \bar{g}^2}} C^{\varphi WB} (\eta_{\mu_3 \mu_5} p_5^{\mu_4} - \eta_{\mu_4 \mu_5} p_5^{\mu_3})$$

A.8 Gluon self interaction vertices



$$-\bar{g}_s f_{a_1 a_2 a_3} (\eta_{\mu_1 \mu_2} p_1^{\mu_3} - \eta_{\mu_1 \mu_2} p_2^{\mu_3} - \eta_{\mu_1 \mu_3} p_1^{\mu_2} + \eta_{\mu_1 \mu_3} p_3^{\mu_2} + \eta_{\mu_2 \mu_3} p_2^{\mu_1} - \eta_{\mu_2 \mu_3} p_3^{\mu_1})$$



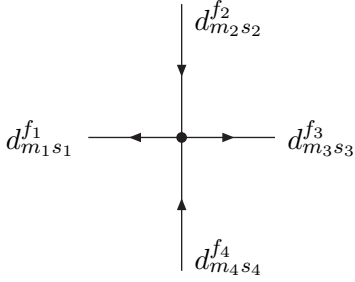
$$\begin{aligned}
& +i\bar{g}_s^2 \left((\eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3} - \eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4}) f_{a_1a_2b_1} f_{a_3a_4b_1} \right. \\
& \quad + (\eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3} - \eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4}) f_{a_1a_3b_1} f_{a_2a_4b_1} \\
& \quad \left. + (\eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} - \eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4}) f_{a_1a_4b_1} f_{a_2a_3b_1} \right)
\end{aligned}$$

A.9 Higgs–gluon vertices

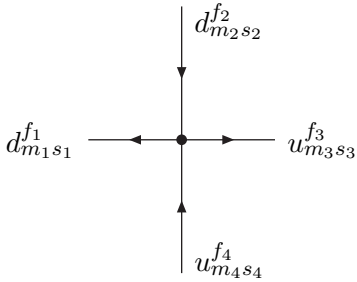
A.10 Four lepton vertices

A.11 Two quark–two lepton vertices

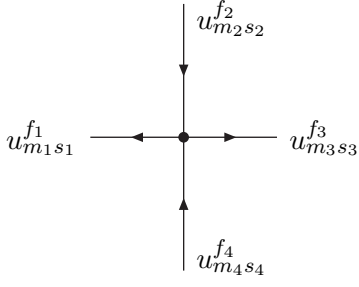
A.12 Four quark vertices



$$\begin{aligned}
& +2i \left(\delta_{m_1 m_4} \delta_{m_2 m_3} C_{f_1 f_4 f_3 f_2}^{dd} (\gamma^\mu P_R)_{s_1 s_4} (\gamma^\mu P_R)_{s_3 s_2} \right. \\
& \quad \left. + \delta_{m_1 m_2} \delta_{m_3 m_4} C_{f_1 f_2 f_3 f_4}^{dd} (\gamma^\mu P_R)_{s_1 s_2} (\gamma^\mu P_R)_{s_3 s_4} \right) \\
& + 2i \left(\delta_{m_1 m_4} \delta_{m_2 m_3} C_{f_1 f_4 f_3 f_2}^{qq1} (\gamma^\mu P_L)_{s_1 s_4} (\gamma^\mu P_L)_{s_3 s_2} \right. \\
& \quad \left. + \delta_{m_1 m_2} \delta_{m_3 m_4} C_{f_1 f_2 f_3 f_4}^{qq1} (\gamma^\mu P_L)_{s_1 s_2} (\gamma^\mu P_L)_{s_3 s_4} \right)
\end{aligned}$$



$$+2i\delta_{m_1 m_2} \delta_{m_3 m_4} K_{f_3 g_2} K_{f_4 g_1}^* (\gamma^\mu P_L)_{s_1 s_2} (\gamma^\mu P_L)_{s_3 s_4} C_{f_1 f_2 g_2 g_1}^{qq1}$$



$$\begin{aligned}
& + 2i K_{f_2 g_1}^* K_{f_4 g_2}^* C_{g_3 g_1 g_4 g_2}^{qq1} (\delta_{m_1 m_4} \delta_{m_2 m_3} K_{f_3 g_3} K_{f_1 g_4} (\gamma^\mu P_L)_{s_1 s_4} (\gamma^\mu P_L)_{s_3 s_2} \\
& + \delta_{m_1 m_2} \delta_{m_3 m_4} K_{f_1 g_3} K_{f_3 g_4} (\gamma^\mu P_L)_{s_1 s_2} (\gamma^\mu P_L)_{s_3 s_4}) \\
& + 2i (\delta_{m_1 m_4} \delta_{m_2 m_3} C_{f_1 f_4 f_3 f_2}^{uu} (\gamma^\mu P_R)_{s_1 s_4} (\gamma^\mu P_R)_{s_3 s_2} \\
& + \delta_{m_1 m_2} \delta_{m_3 m_4} C_{f_1 f_2 f_3 f_4}^{uu} (\gamma^\mu P_R)_{s_1 s_2} (\gamma^\mu P_R)_{s_3 s_4})
\end{aligned}$$

A.13 Baryon and lepton number violating four fermion vertices